String Reorder

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1743 | Introductory Problems | https://cses.fi/problemset/task/1743

Your task is to reorder the characters of a string so that no two adjacent characters are the same. What is the lexicographically minimal such string?

Input

The only line has a string of length n consisting of characters A–Z.

Output

Print the lexicographically minimal reordered string where no two adjacent characters are the same. If it is not possible to create such a string, print -1.

Constraints

- $1 \le n \le 10^6$

Example

Input
HATTIVATTI

Output
AHATITITVT